

User Manual

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Introduction

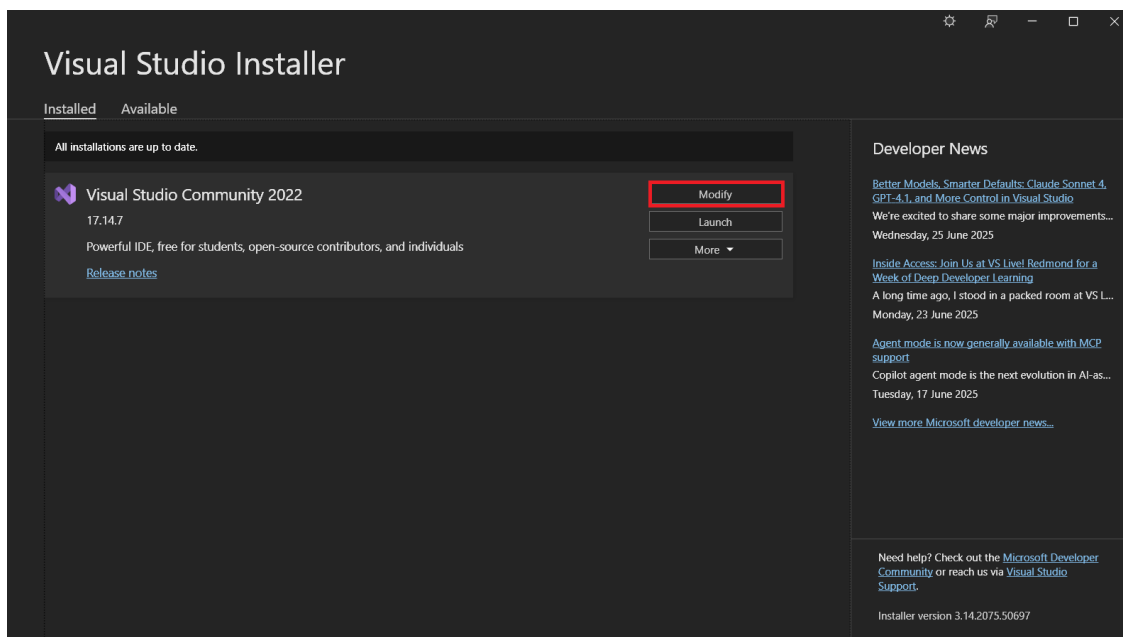
After getting all the necessary files and library, this guide will show you how to use the app from installing the required **C++ components** in **Visual Studio** to performing basic mathematical operations.

Installing C++ Components in Visual Studio

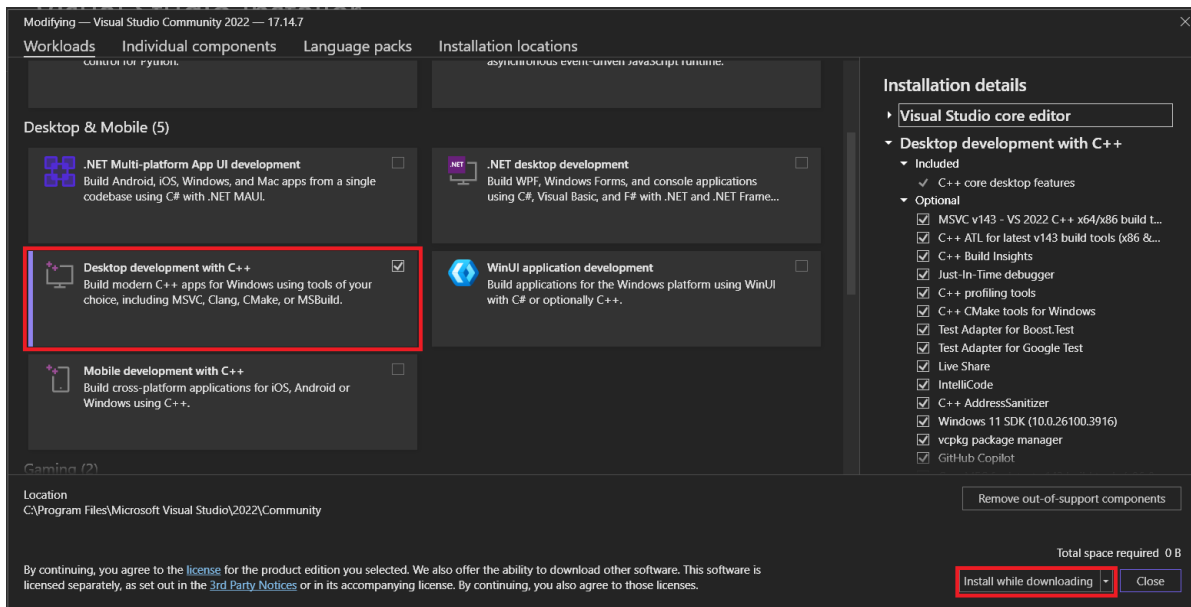
The software of choice here is **Visual Studio**.

Steps:

1. Open **Visual Studio installer**
2. Click **Modify**



- Click **Desktop development with C++** as this will install the necessary components and click **Install while downloading**

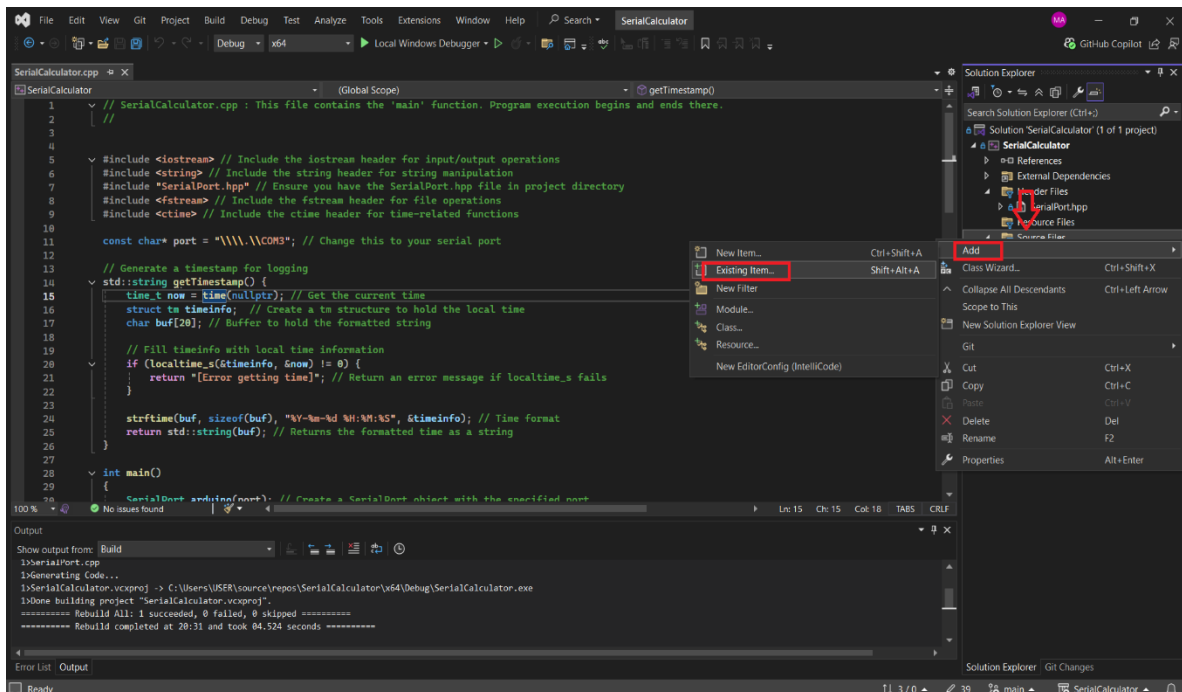


Importing SerialPort Library (Console or Visual Studio)

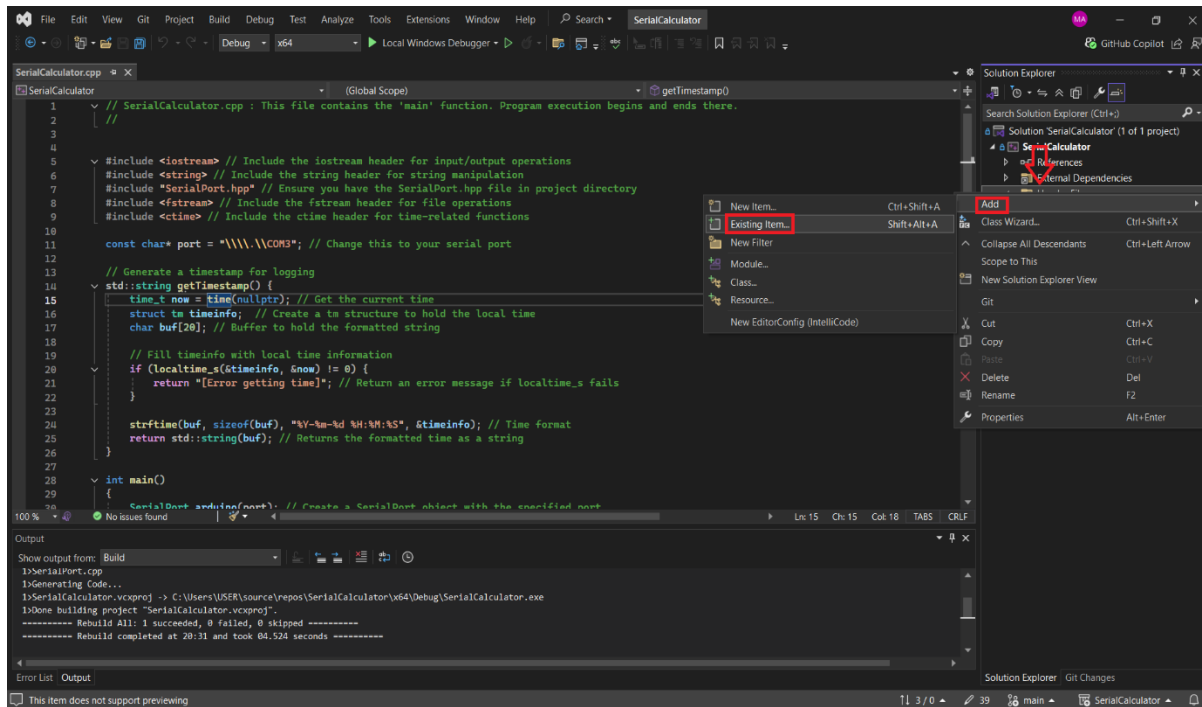
This can be done in two ways:

1. Import in Visual Studio

- Importing **SerialPort.cpp**



- Importing header file: **SerialPort.hpp**

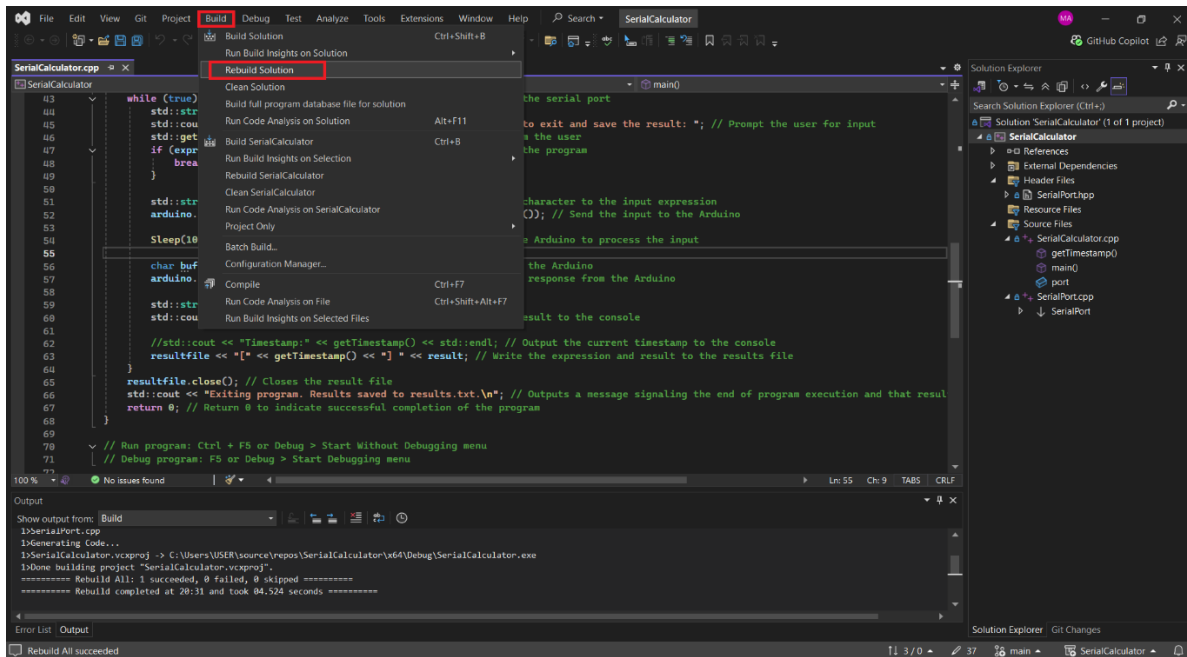


2. Extracting the **SerialPort.zip**, copy and paste the **SerialPort.cpp** and **SerialPort.hpp** directly into the **project folder**

Building the Calculator

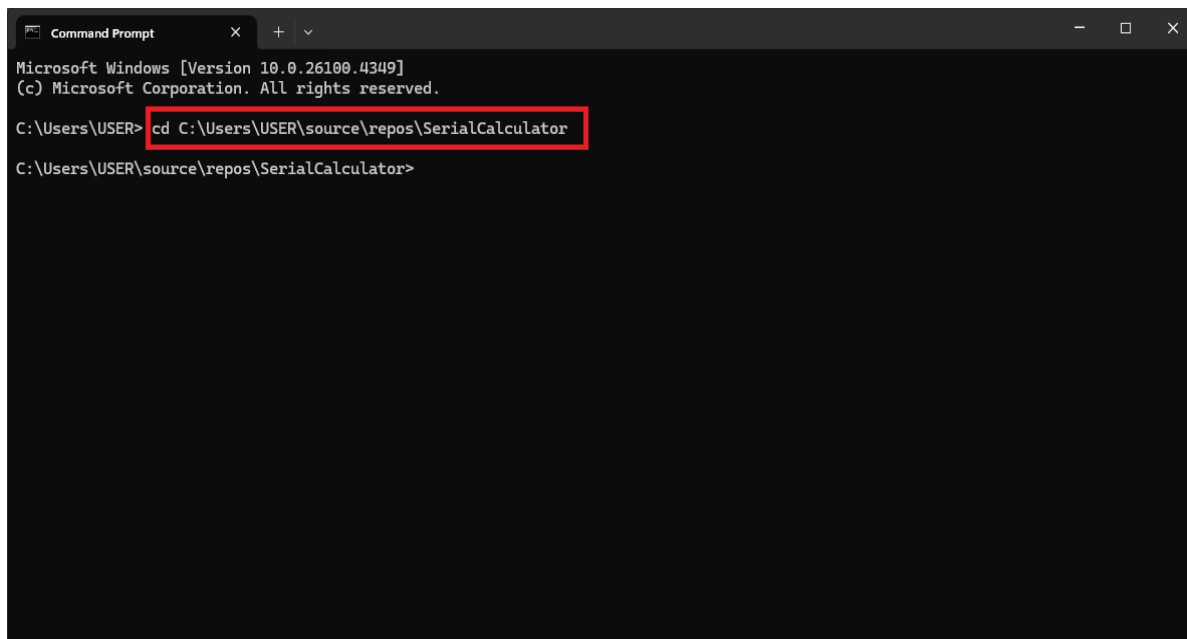
Two methods can be used to do this, either through Visual Studio or the Console App:

1. Visual Studio

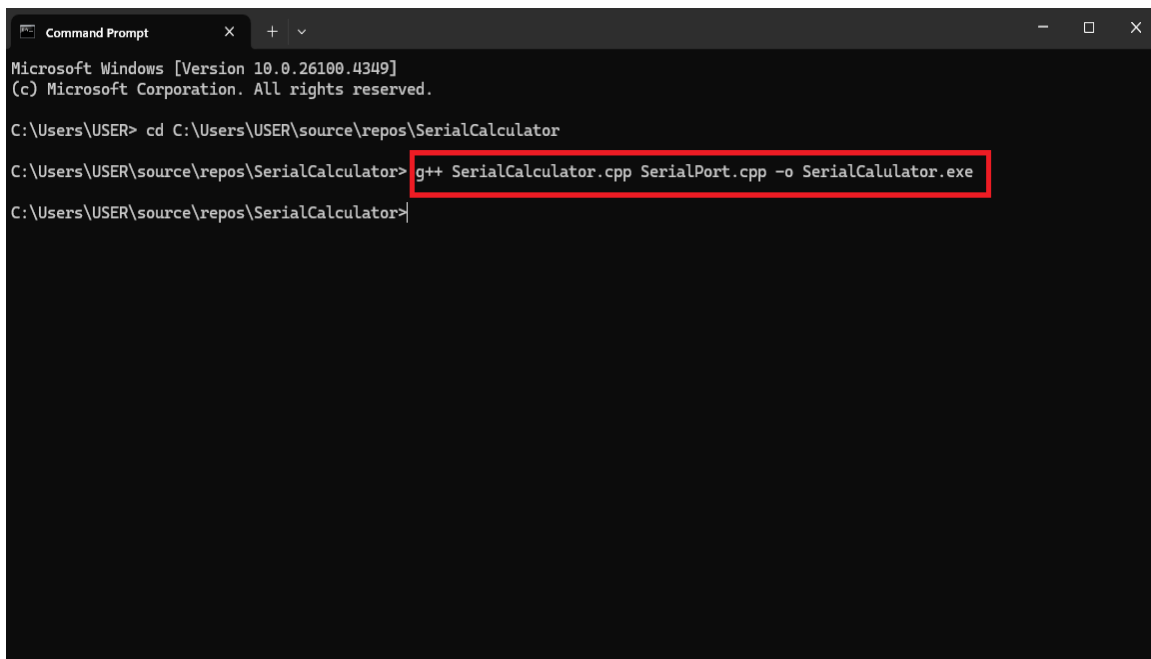


2. Console App

- Change **directory** to the **project directory**



- Creating **executable (.exe)** file



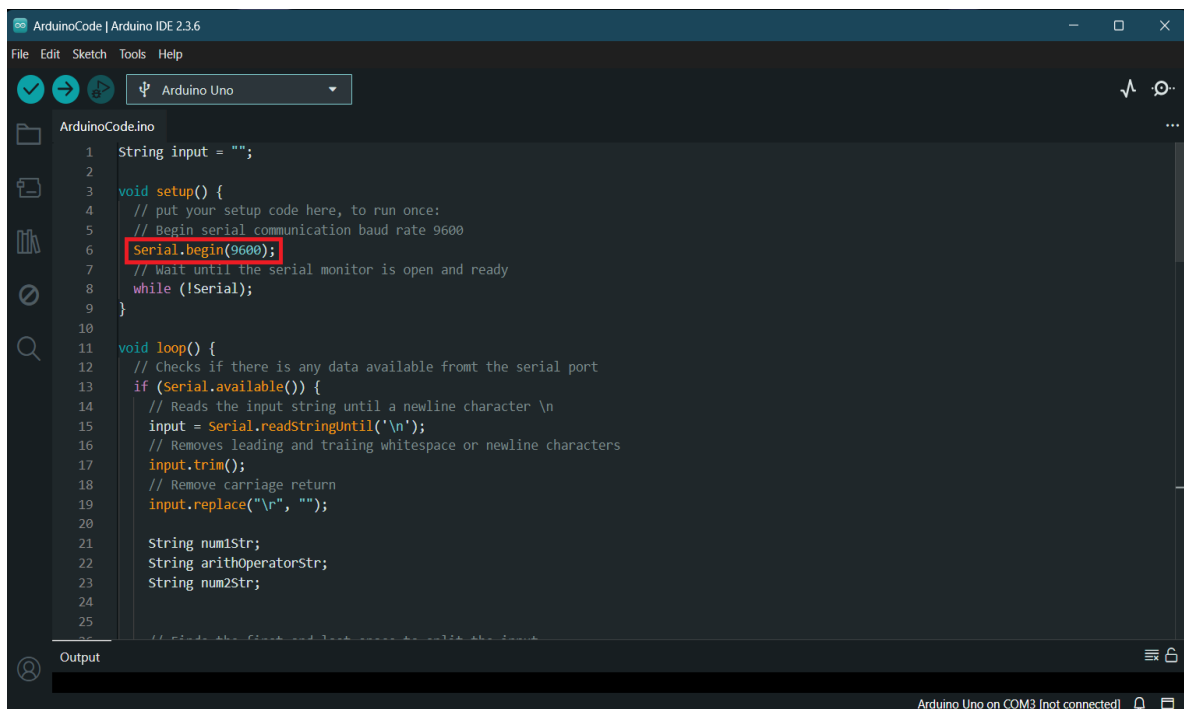
```
Microsoft Windows [Version 10.0.26100.4349]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER> cd C:\Users\USER\source\repos\SerialCalculator

C:\Users\USER\source\repos\SerialCalculator> g++ SerialCalculator.cpp SerialPort.cpp -o SerialCalculator.exe

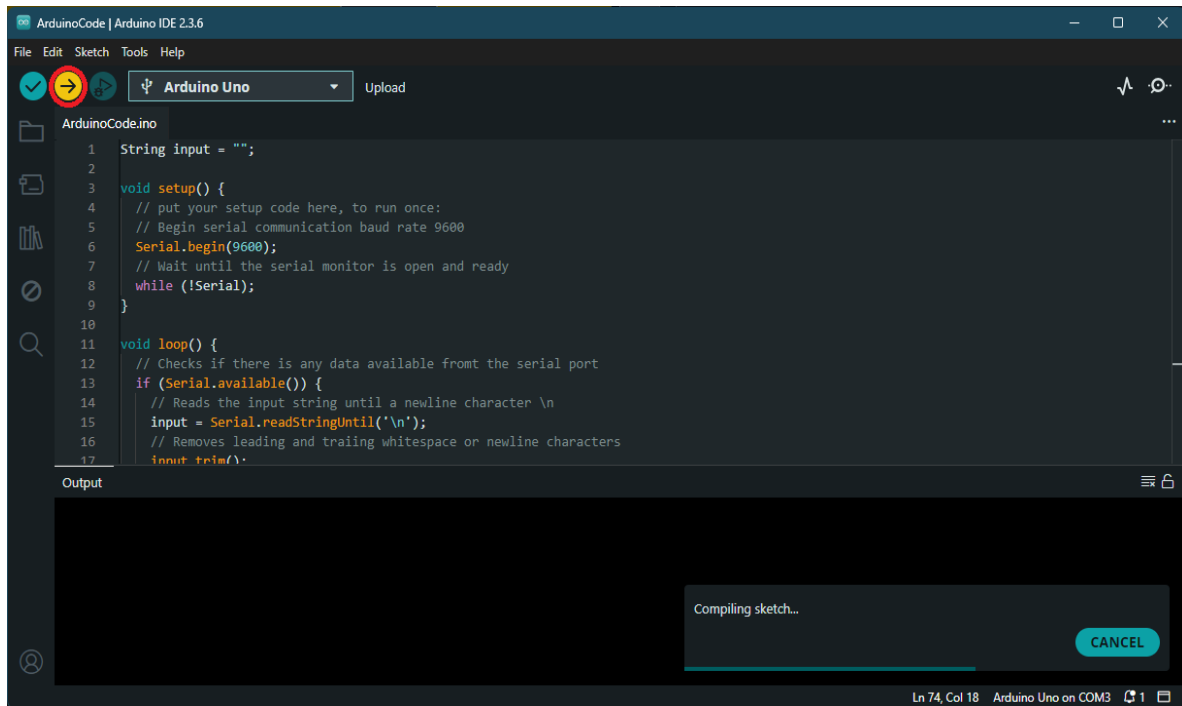
C:\Users\USER\source\repos\SerialCalculator>
```

Setting Up the Connection — baud rate



```
ArduinoCode.ino
1 String input = "";
2
3 void setup() {
4   // put your setup code here, to run once:
5   // Begin serial communication baud rate 9600
6   Serial.begin(9600);
7   // Wait until the serial monitor is open and ready
8   while (!Serial);
9 }
10
11 void loop() {
12   // Checks if there is any data available from the serial port
13   if (Serial.available()) {
14     // Reads the input string until a newline character \n
15     input = Serial.readStringUntil('\n');
16     // Removes leading and trailing whitespace or newline characters
17     input.trim();
18     // Remove carriage return
19     input.replace("\r", "");
20
21     String num1Str;
22     String arithOperatorStr;
23     String num2Str;
24
25     // Reads the first and last spaces to split the input
```

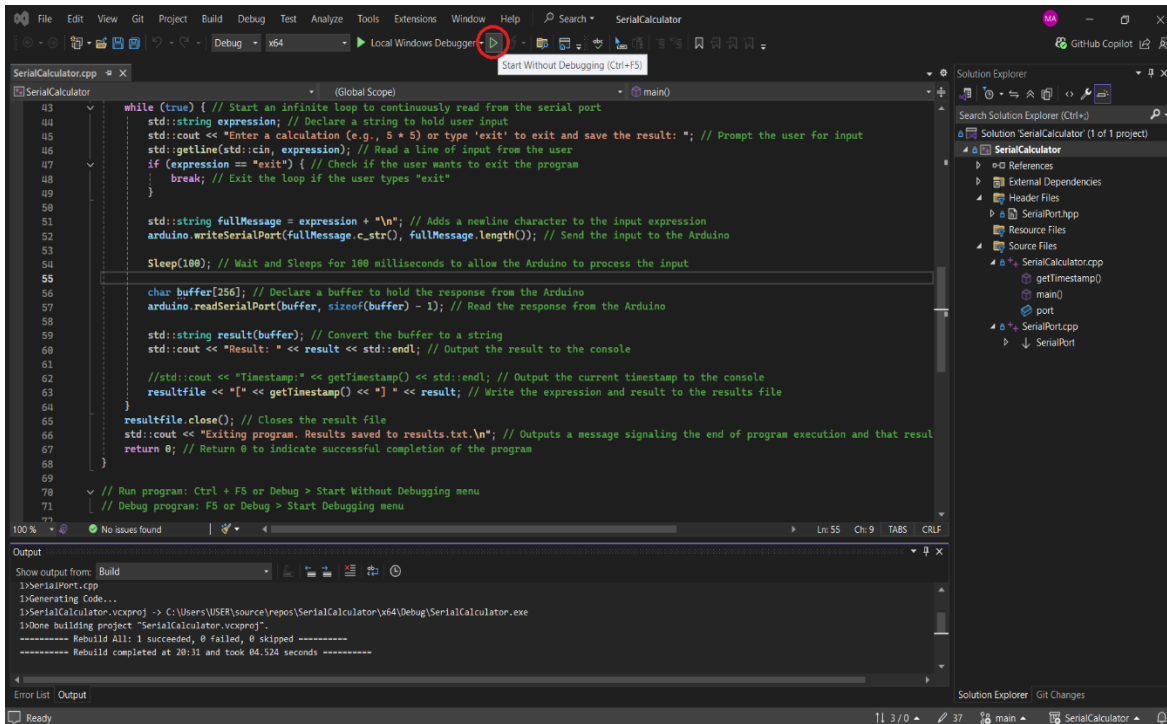
Uploading *ArduinoCode.ino* to Microcontroller



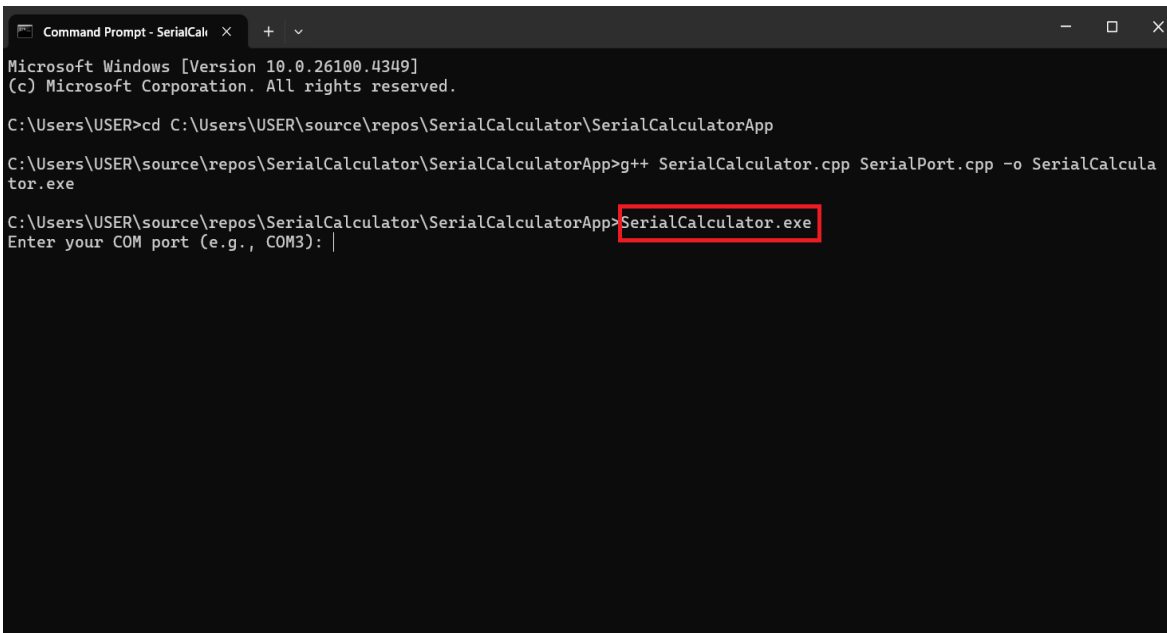
Running the App — via Visual Studio | Console | .exe file

The app can be run through three different ways:

1. Visual Studio



2. Console app

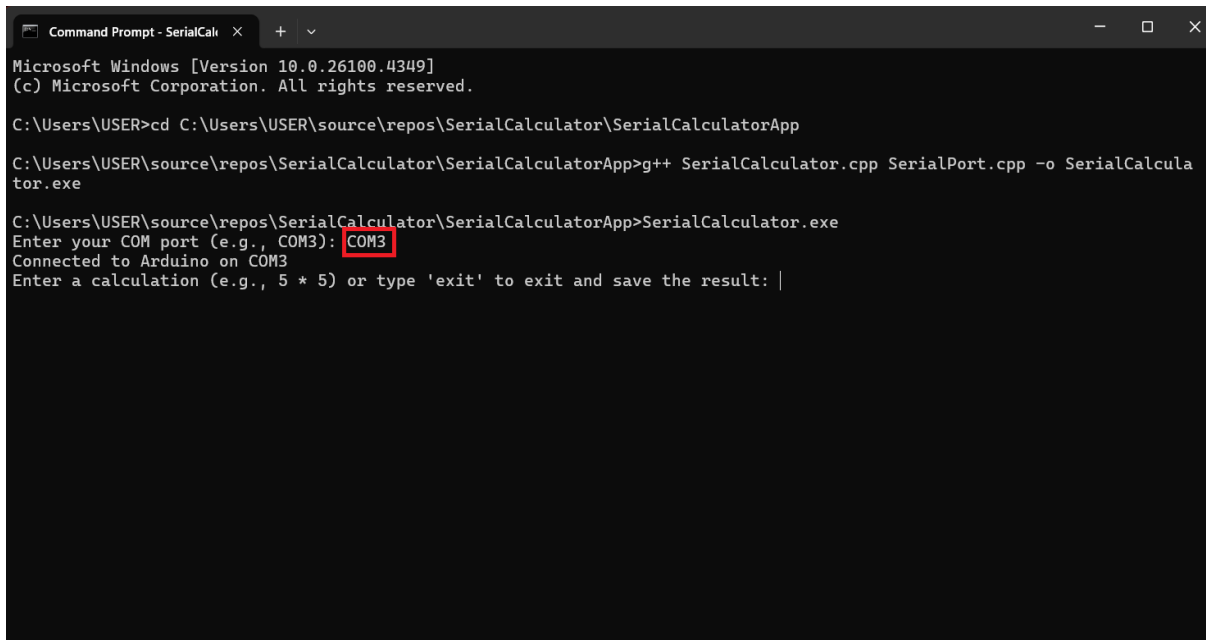


3. Project folder



Selecting the COM port

Inputting the **COM port**: After opening the **.exe file**, a prompt is shown where the user is requested to type the **COM port**. Type the **COM port** (Style: *COM3* or *com3*) assigned by your **PC** to the **microcontroller** (Open **Arduino IDE** or **device manager** on Windows to check if unsure)



```
Command Prompt - SerialCalc
Microsoft Windows [Version 10.0.26100.4349]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER>cd C:\Users\USER\source\repos\SerialCalculator\SerialCalculatorApp

C:\Users\USER\source\repos\SerialCalculator\SerialCalculatorApp>g++ SerialCalculator.cpp SerialPort.cpp -o SerialCalculator.exe

C:\Users\USER\source\repos\SerialCalculator\SerialCalculatorApp>SerialCalculator.exe
Enter your COM port (e.g., COM3): COM3
Connected to Arduino on COM3
Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: |
```

Entering a Mathematical Expression

Typically, there are two ways for the expression to be entered that would be considered valid by the microcontroller and it is of this format: `<number1> <operator> <number2>`

1. Sample entry 1

```
Command Prompt - SerialCalci x + v
Microsoft Windows [Version 10.0.26100.4349]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER>cd C:\Users\USER\source\repos\SerialCalculator\SerialCalculatorApp

C:\Users\USER\source\repos\SerialCalculator\SerialCalculatorApp>g++ SerialCalculator.cpp SerialPort.cpp -o SerialCalculator.exe

C:\Users\USER\source\repos\SerialCalculator\SerialCalculatorApp>SerialCalculator.exe
Enter your COM port (e.g., COM3): COM3
Connected to Arduino on COM3
Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 9 * 5
Result: 9 * 5 = 45

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: |
```

2. Sample entry 2

```
Command Prompt - SerialCalci x + v
Microsoft Windows [Version 10.0.26100.4349]
Microsoft Corporation. All rights reserved.

Users\USER>cd C:\Users\USER\source\repos\SerialCalculator\SerialCalculatorApp

Users\USER\source\repos\SerialCalculator\SerialCalculatorApp>g++ SerialCalculator.cpp SerialPort.cpp -o SerialCalculator.exe

Users\USER\source\repos\SerialCalculator\SerialCalculatorApp>SerialCalculator.exe
Enter your COM port (e.g., COM3): COM3
Connected to Arduino on COM3
Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 9 * 5
Result: 9 * 5 = 45

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 9*5
Result: 9 * 5 = 45

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: |
```

Supported Operations

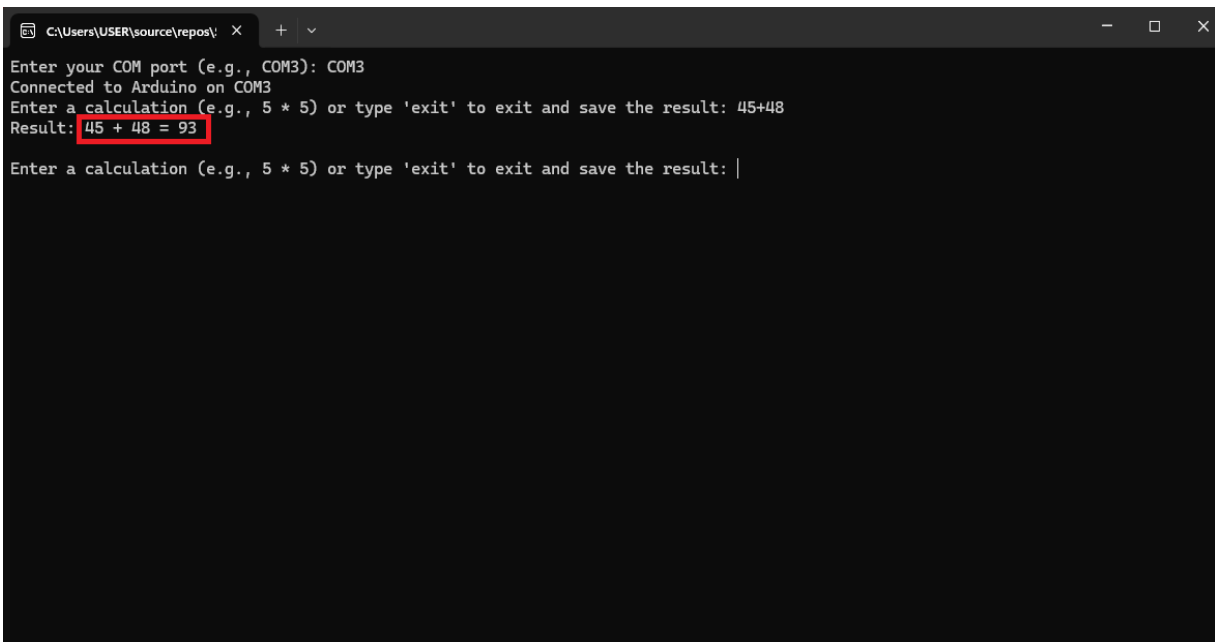
The following list contains supported operations and an example of their usage:

- Addition (+)
- Subtraction (-)
- Multiplication (*)

- Division (/)
- Power (^)
- Root ($n\sqrt{}$)
- Remainder (%)

Example of Usage:

1. Addition (+)



```
C:\Users\USER\source\repos\ > x + v
Enter your COM port (e.g., COM3): COM3
Connected to Arduino on COM3
Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 45+48
Result: 45 + 48 = 93
Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: |
```

2. Subtraction (-)

```
C:\Users\USER\source\repos\ X + v
Enter your COM port (e.g., COM3): COM3
Connected to Arduino on COM3
Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 45+48
Result: 45 + 48 = 93

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 60-28
Result: 60 - 28 = 32

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result:
```

3. Multiplication (*)

```
C:\Users\USER\source\repos\ X + v
Enter your COM port (e.g., COM3): COM3
Connected to Arduino on COM3
Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 45+48
Result: 45 + 48 = 93

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 60-28
Result: 60 - 28 = 32

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 5*9
Result: 5 * 9 = 45

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: |
```

4. Division (/)

```
C:\Users\USER\source\repos\ X + v
Enter your COM port (e.g., COM3): COM3
Connected to Arduino on COM3
Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 45+48
Result: 45 + 48 = 93

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 60-28
Result: 60 - 28 = 32

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 5*9
Result: 5 * 9 = 45

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 67/7
Result: 67 / 7 = 9.5714

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result:
```

5. Power (^)

```
C:\Users\USER\source\repos\ X + v
Enter your COM port (e.g., COM3): COM3
Connected to Arduino on COM3
Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 45+48
Result: 45 + 48 = 93

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 60-28
Result: 60 - 28 = 32

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 5*9
Result: 5 * 9 = 45

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 67/7
Result: 67 / 7 = 9.5714

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 5^4
Result: 5 ^ 4 = 625

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: |
```

6. Root ($n\sqrt{}$): The root is entered in this format (`<n-th root> <r> <value>`). For example, taking the cubic root of 729 will look like: `3 r 729`.

```
C:\Users\USER\source\repos\ X + v - □ X
Enter your COM port (e.g., COM3): COM3
Connected to Arduino on COM3
Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 45+48
Result: 45 + 48 = 93

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 60-28
Result: 60 - 28 = 32

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 5*9
Result: 5 * 9 = 45

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 67/7
Result: 67 / 7 = 9.5714

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 5^4
Result: 5 ^ 4 = 625

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 5r3125
Result: 5 r 3125 = 5

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: |
```

7. Remainder (%): This gives the remainder of the **dividend/divisor** in the form:

$$200 \% 3 = 2$$

```
C:\Users\USER\source\repos\ X + v - □ X
Enter your COM port (e.g., COM3): COM3
Connected to Arduino on COM3
Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 45+48
Result: 45 + 48 = 93

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 60-28
Result: 60 - 28 = 32

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 5*9
Result: 5 * 9 = 45

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 67/7
Result: 67 / 7 = 9.5714

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 5^4
Result: 5 ^ 4 = 625

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 5r3125
Result: 5 r 3125 = 5

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 208%5
Result: 208 % 5 = 3

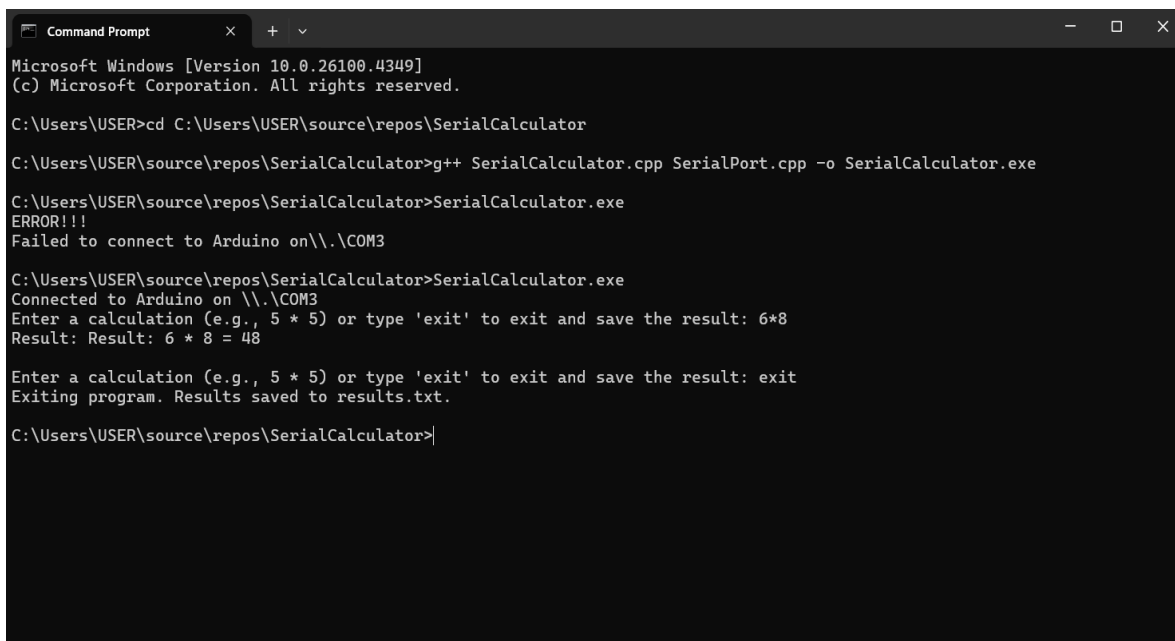
Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: |
```

Saving the Result

Once the user has completed the desired operations, to save the **results** from the calculations, the user must type the command '**exit**' to indicate to the program that the set of operation is complete. It then saves the operations to a **results.txt** files in **the project folder**.

Note: Abruptly **closing** the **command window/Console App** would result in **failure** of the program in **saving** the previous operations to the **file**.

All **previous operations** are shown in the file.



```
Microsoft Windows [Version 10.0.26100.4349]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER>cd C:\Users\USER\source\repos\SerialCalculator

C:\Users\USER\source\repos\SerialCalculator>g++ SerialCalculator.cpp SerialPort.cpp -o SerialCalculator.exe

C:\Users\USER\source\repos\SerialCalculator>SerialCalculator.exe
ERROR!!!
Failed to connect to Arduino on \\.\COM3

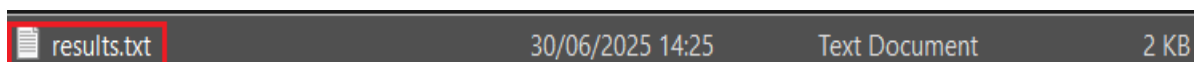
C:\Users\USER\source\repos\SerialCalculator>SerialCalculator.exe
Connected to Arduino on \\.\COM3
Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 6*8
Result: Result: 6 * 8 = 48

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: exit
Exiting program. Results saved to results.txt.

C:\Users\USER\source\repos\SerialCalculator>
```

Checking the **result.txt** File Location

To check the **history** of operations previously performed, go to the **project folder** and find the **results.txt** file. You can open the file with any compatible software including the native **notebook** software that comes pre-installed with **Windows**.



Error Messages

- **“Error: Division by zero”** – Division by zero attempted

- ***“Error: Invalid operator”*** – Unsupported operator used
- ***“Error: No valid operator found”*** – No operator detected in input
- ***“Error: Invalid input format”*** – Incorrect input string

Troubleshooting

Common Issues

“Failed to connect to Arduino”

- **Cause:** Incorrect **COM** port, Arduino not connected or Serial monitor still open in Arduino IDE
- **Solution:**
 - Close the serial monitor in Arduino IDE
 - Ensure corrected port is selected by updating port variable in code (this can be verified via the Arduino IDE or Device Manager)
 - Ensure Arduino is connected
 - Ensure the proper drivers are installed, if not, check Device Manager

“Could not open results file”

- **Cause:** File permissions or disk space issues
- **Solution:**
 - Check available disk space
 - Ensure directory is writable
 - Run application as administrator

Arduino not responding

- **Cause:** Arduino not programmed or serial connection issues
- **Solution:**
 - Verify Arduino firmware is uploaded
 - Check serial monitor in Arduino IDE
 - Try different USB cable or port

Incorrect Calculation

- **Cause:** Input parsing issues or precision limitations
- **Solution:**
 - Verify input format matches supported patterns (Format: `<number1> <operator> <number2>`)
 - Check for extra spaces or characters
 - Understand floating-point precision limits

Shows it has connected but not showing that next line, for example: “Connected to Arduino on COM3”

- **Cause:** Incorrect **COM port** typed in, could be connected to a port (Such as the one assigned to Bluetooth) but not the one assigned to the Arduino by the PC
- **Solution:**
 - Go to **device manager**, check the Arduino Port (Unplug and plug back in if unsure) and close the app and restart. Then type in the appropriate port